

What computational methods have been used to predict novel biomedical associations from knowledge graphs, and how have link prediction or missing edge inference approaches been validated?

Computational methods for predicting biomedical associations range from knowledge graph embeddings to advanced Graph Neural Networks and hybrid models, and are predominantly validated using computational metrics on split datasets, with experimental or clinical confirmation being notably rare.

Abstract

A wide range of computational methods are employed to predict novel biomedical associations from knowledge graphs, with a notable shift from traditional knowledge graph embedding (KGE) models like TransE and ComplEx towards more advanced architectures. Graph Neural Networks (GNNs), particularly those incorporating path-based reasoning and attention mechanisms, are increasingly utilized to capture complex topological patterns inherent in biomedical data. Hybrid and ensemble approaches are also prevalent, combining multiple model types or integrating multimodal data, such as textual information from large language models, to enhance predictive accuracy. Neuro-symbolic methods that fuse neural networks with logical rules are emerging to improve reasoning and provide the interpretability crucial for biomedical applications. These methods are applied to diverse prediction tasks, primarily drug repurposing, drug-target interaction prediction, and gene-disease association discovery.

Validation of these link prediction approaches predominantly relies on computational metrics. Standard practices include splitting data into training and test sets and using k-fold cross-validation, with performance assessed by ranking metrics like Mean Reciprocal Rank (MRR) and Hits@k, alongside classification metrics such as AUC-ROC and F1-score. A growing number of studies are adopting more rigorous validation strategies, including temporal splits and "cold-start" scenarios, to better assess generalizability and mitigate data leakage. However, a significant gap persists between computational assessment and real-world verification. While many models report high performance on computational benchmarks, only a minority of studies validate their novel predictions through external evidence such as literature reviews, independent clinical or biological databases, or direct experimental confirmation. This highlights a critical limitation in translating computational hypotheses into biologically validated knowledge.

Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "What computational methods have been used to predict novel biomedical associations from knowledge graphs, and how have link prediction or missing edge inference approaches been validated?"

The search returned 500 total results from Elicit.

We retrieved 500 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Link Prediction Methods:** Does this study apply computational link prediction or missing edge inference methods to biomedical knowledge graphs?

- **Biomedical Association Prediction:** Is this study focused on predicting novel biomedical associations (e.g., drug-target, gene-disease, protein-protein, drug-drug interactions)?
- **Method Validation:** Does this study include validation or evaluation of their prediction methods?
- **Knowledge Graph Data Sources:** Does this study use knowledge graphs, heterogeneous networks, or multi-relational biomedical databases as primary data sources?
- **Appropriate Study Type:** Is this study an original research article, systematic review, meta-analysis, or conference paper describing computational methodologies?
- **Beyond Graph Construction Only:** Does this study go beyond solely focusing on knowledge graph construction, curation, or visualization to include link prediction?
- **Graph Structure Utilization:** Does this study utilize graph structure or relationships rather than being limited to traditional machine learning on tabular data?
- **Biomedical Domain Focus:** Is this study focused on biomedical domains rather than exclusively on non-biomedical domains (e.g., social networks, recommendation systems)?
- **Substantial Research Content:** Is this study a substantial research contribution rather than an editorial comment, letter, opinion piece, or case report?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Computational Method:**

Extract the core computational approach used for predicting biomedical associations from knowledge graphs, including:

- Primary algorithm or method name (e.g., TransE, ensemble learning, neural Bellman-Ford)
- Type of approach (knowledge graph embedding, graph neural network, ensemble, etc.)
- Key technical innovations or modifications to standard methods
- Whether the method is novel or an adaptation of existing approaches

- **Biomedical Application:**

Extract details about what biomedical associations were predicted using link prediction, including:

- Type of associations (drug-target interactions, gene-disease, drug-disease, gene function, synthetic lethality, etc.)
- Specific entities involved (genes, proteins, drugs, diseases, pathways, etc.)
- Clinical or biological relevance of the predicted associations
- Whether predictions focused on novel discovery or validation of known associations

- **Knowledge Graph Details:**

Extract information about the knowledge graph(s) used for link prediction, including:

- Dataset name and source (PubMed KG, DrugBank, etc.)
- Graph size (number of nodes, edges, relations)
- Types of entities and relations included
- Whether custom knowledge graphs were constructed or existing ones were used

- Data preprocessing or graph modification steps

- **Validation Strategy:**

Extract comprehensive details about how the link prediction approach was validated, including:

- Computational validation methods (cross-validation, train/test splits, negative sampling strategies)
- Evaluation metrics used (MRR, Hits@k, AUC-ROC, precision-recall, etc.)
- Benchmark datasets or gold standards used for evaluation
- Comparison to baseline methods or state-of-the-art approaches
- Whether biological or clinical validation was performed beyond computational metrics

- **Performance Results:**

Extract quantitative performance results and novel discoveries, including:

- Specific metric values achieved (MRR, Hits@k scores, AUC values, etc.)
- Performance comparison to baseline or competing methods
- Statistical significance of improvements
- Novel biomedical associations discovered or predicted
- Validation of predictions through literature, databases, or experimental confirmation

- **Technical Implementation:**

Extract technical implementation details relevant to reproducing or comparing methods, including:

- Model architecture specifics (embedding dimensions, neural network layers, etc.)
- Training procedures and hyperparameters
- Computational requirements and scalability considerations
- Software frameworks or tools used
- Handling of knowledge graph heterogeneity or multi-relational aspects

An abstract is not available for this report.

Study Characteristics

A comprehensive overview of the included studies reveals a focus on developing and applying computational methods for link prediction within biomedical knowledge graphs. The primary objective across most studies is the inference of novel biomedical associations, such as drug-target interactions, gene-disease relationships, and drug repurposing candidates. Many studies introduce novel computational methods or significant adaptations of existing ones to address the specific challenges of biomedical data, including noise, sparsity, and heterogeneity. The biomedical domains are diverse, covering pharmacology, genetics, and clinical medicine, with applications aimed at accelerating drug discovery, understanding disease mechanisms, and enabling personalized medicine. Research is predominantly conducted in academic settings, often involving the construction of custom knowledge graphs from publicly available databases or the use of established benchmark datasets. A significant portion of the literature focuses on method development, proposing new algorithms for knowledge graph embedding, graph neural networks, and ensemble approaches, while also applying these methods to solve concrete biomedical problems.

Citation	Year	Primary Objective	Biomedical Domain	Novel Method or Application
Yue Hu et al.	2024	Enhance link prediction through path-based reasoning	Drug discovery, personalized medicine	Novel method (BioPathNet)
Yue Hu et al.	2024a	Apply path-based reasoning to biomedical knowledge graphs	Drug-disease interactions, synthetic lethality	Novel adaptation of NBFNet
Joana Vilela et al.	2022	Predict disease-gene associations using KGE methods	Personalized medicine, neurology (ASD)	Adaptation of existing methods
Haji Gul et al.	2025	Improve drug target discovery with multi-context aware sampling	Drug discovery, systems biology	Novel adaptation of MuCo-KGC
Tengfei Ma et al.	2023	Denoise biomedical knowledge graphs for robust interaction prediction	Drug discovery, therapeutics	Novel framework (DenoisedLP)
Meng Wang et al.	2021	Predict rich DDI information via KG and text embedding	Clinical pharmacology	Novel framework (Predicting Rich DDI)
Yongkang Xiao et al.	2024	Enhance GNN-based link prediction with LLM embeddings and domain knowledge	Drug repurposing	Novel method (FuseLinker)
Yahui Long et al.	2022	Pre-train graph neural networks for link prediction in biomedical networks	Drug discovery, disease pathology	Novel framework (PT-GNN)
Zhenxiang Gao et al.	2022	Develop a knowledge graph framework for drug repurposing	Drug repurposing, neurology (Alzheimer's)	Novel framework (KG-Predict)
V. Nováček et al.	2020	Predict polypharmacy side-effects using a new KGE technique	Clinical pharmacology	Novel method

Citation	Year	Primary Objective	Biomedical Domain	Novel Method or Application
Chih-Hsu Lin et al.	2018	Predict future disease-gene-chemical associations using multimodal network diffusion	Precision medicine	Adaptation of existing methods
Yaqing Wang et al.	2023	Develop an accurate and interpretable method for DDI prediction	Drug safety, biomedicine	Novel adaptation (KnowDDI)
M. Chang et al.	2024	Predict disease-gene/protein associations using a multi-head attention mechanism	Precision medicine, drug target identification	Novel method (CMEI)
M. O. Faruk et al.	2025	Develop a large-scale pharmacogenomic KG for discovery	Pharmacogenomics, precision therapeutics	Adaptation of existing methods
Nansu Zong et al.	2019	Predict drug-target associations using heterogeneous network mining	Drug discovery, disease treatment	Adaptation of existing methods
Lizzy Farrugia et al.	2023	Predict drug-drug interactions using knowledge graphs	Pharmacology, drug safety	Adaptation of existing methods
F. Aisopos et al.	2023	Compare methods for drug-gene interaction prediction on a literature KG	Drug discovery, drug repurposing	Adaptation of existing methods
Catarina Canastra et al.	2025	Systematically evaluate KGEs for gene-disease association prediction	Disease identification, drug repurposing	Novel comparison framework
Yang Zhao et al.	2024	Enhance link prediction using community knowledge graph abstraction	Biomedical knowledge discovery	Adaptation of existing methods

Citation	Year	Primary Objective	Biomedical Domain	Novel Method or Application
Kevin McCoy et al.	2021	Apply biomedical text link prediction for drug discovery with a COVID-19 case study	Drug discovery, infectious diseases	Adaptation of existing methods
Yushan Liu et al.	2021	Integrate logical rules into neural multi-hop reasoning for drug repurposing	Drug repurposing	Novel method (PoLo)
Vignesh Prabhakar et al.	2023	Use an ensemble learning approach for link prediction on large-scale biomedical KGs	Drug repurposing and discovery	Adaptation of existing methods
Roger Tu et al.	2024	Repurpose drugs using consilience of KG completion methods	Drug repurposing	Adaptation of existing methods
Sameh K. Mohamed et al.	2019	Discover protein drug targets using knowledge graph embeddings	Drug discovery	Novel method (TriModel)
Gamal K. O. Crichton et al.	2018	Evaluate neural networks for link prediction in realistic biomedical graphs	DTI, PPI, Literature-Based Discovery	Adaptation of existing methods
Jinjiang Guo et al.	2021	Predict biomedical network links using a heterogeneous graph based deep learning model	Biomedical interaction discovery	Novel method (GPLP)
Budhaditya Saha et al.	2020	Predict missing and noisy links in a clinical knowledge base	Clinical knowledge management	Novel application
Sameh K. Mohamed et al.	2019a	Discover drug targets using a customized KGE model	Drug discovery	Novel adaptation of ComplEx

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Nansu Zong et al.	2019a	Predict drug-targets utilizing heterogeneous bio-linked network embeddings	Drug discovery, disease treatment	Adaptation of existing methods
Tien Dang et al.	2025	Enhance link prediction through multimodal contrastive representation learning	Drug-target interaction prediction	Novel method
Mona Alshahrani et al.	2021	Apply and evaluate knowledge graph embeddings in biomedical data	Data analytics, machine learning model building	Adaptation of existing methods
Xiaomin Liang et al.	2019	Predict biomedical relationships using a knowledge and graph embedding cascade model	Biomedical network analysis	Adaptation with novel framework
U. M. Singh-Blom et al.	2013	Predict gene-disease associations using methods inspired by social network analyses	Genomics, disease biology	Adaptations of existing approaches
Wenhui Wang et al.	2012	Predict drug targets based on heterogeneous graph inference	Drug development	Novel adaptation for heterogeneous graphs
Gokhan Bakal et al.	2018	Predict treatment and causative relations using semantic patterns over biomedical KGs	Biomedical relation prediction	Novel application of graph patterns
A. Daowd et al.	2022	Apply a KGC method to literature-based discovery for cancer drug repurposing	Cancer drug repurposing	Adaptation of existing methods

Citation	Year	Primary Objective	Biomedical Domain	Novel Method or Application
D. Bean et al.	2017	Predict unknown adverse drug reactions using a knowledge graph and validate in EHRs	Pharmacovigilance, patient safety	Adaptation of machine learning approaches
Mona Alshahrani et al.	2022	Combine biomedical KGs and text to improve predictions for drug-target interactions	Drug repurposing	Adaptation with innovative data integration
P. Sevon et al.	2006	Discover links in graphs derived from biological databases	Biological hypothesis generation	Novel application of reliability measures
Meghamala Sinha et al.	2024	Improve KG completion performance using a weighted ensemble approach	Drug discovery	Novel method (WeightedKgBlend)
R. Çelebi et al.	2019	Evaluate KGE approaches for DDI prediction in realistic settings	Drug safety, translational research	Adaptation of existing methods
Ankit Patel et al.	2021	Find novel links in a COVID-19 knowledge graph using graph embedding techniques	Literature search and discovery	Adaptation of existing methods
Jake Lever et al.	2018	Apply a collaborative filtering approach to biomedical knowledge discovery	Literature-based discovery	Adaptation of SVD
Awais Bajwa et al.	2019	Analyze interaction networks using semantic similarity based on translation embeddings	Drug-target interaction discovery	Adaptation of TransE (SimTransE)

Citation	Year	Primary Objective	Biomedical Domain	Novel Method or Application
Olawumi Olasunkanmi et al.	2024	Propose an explainable enrichment-driven reasoner for large KGs in drug repurposing	Drug repurposing, Alzheimer's disease	Novel method (EDGAR)
Alejandro Flores et al.	2015	Predict potential novel links from dense subgraphs by exploiting semantics	Drug-target interaction prediction	Adaptation of existing methods
Susana Nunes et al.	2023	Predict gene-disease associations using multi-domain knowledge graph embeddings	Diagnostic marker and therapeutic target identification	Adaptation of existing methods to multiple ontologies
Aryo Pradipta Gema et al.	2023	Evaluate the usefulness of KGEs in the biomedical domain for various tasks	Link prediction, rule learning, polypharmacy tasks	Adaptation of existing methods
OUP accepted manuscript	2022	Present a software suite for predicting, explaining, and exploring links in biomedical KGs	Hypothesis generation	Novel method (SAFRAN)
V. Nováček et al.	2019	Accurately predict kinase-substrate networks using knowledge graphs	Phosphoproteomics, cellular signaling	Adaptation of ComplEx
K. Bougiatiotis et al.	2020	Predict DDI on a biomedical literature knowledge graph	Drug safety, disease management	Novel adaptation of graph embedding techniques
Mona Alshahrani et al.	2016	Develop neuro-symbolic representation learning on biological knowledge graphs	Function prediction, candidate gene finding, PPI prediction	Adaptation of existing methods with novel integration

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Binon Teji et al.	2024	Empirically evaluate graph embedding techniques for predicting missing links in biological networks	Network biology	Adaptation of existing methods
Ozge Gurbuz et al.	2022	Develop an explainable target-disease prediction method for indication expansion	Drug discovery, indication expansion	Adaptation of existing methods
Xiang Yue et al.	2019	Evaluate graph embedding on biomedical networks for various applications	DDA, DDI, PPI prediction; node classification	Adaptations of existing approaches
Daniel N. Sosa et al.	2024	Study the semantics-topology trade-off for knowledge inference-based pharmacological discovery	Pharmacological discovery, drug repurposing	Adaptation of existing methods
Daniel Daza et al.	2023	Develop a modular framework for learning on multimodal biomedical knowledge graphs	Drug-protein interaction prediction, drug discovery	Novel method (BioBLP)
Francesco Gualdi et al.	2024	Predict gene disease associations for diseases with curtailed information using KGE	Disease biology, rare disease genetics	Novel methods (DLemb, BioKG2Vec)
Ziyan Zhang et al.	2020	Predict links in biomedical bipartite networks using a graph regularized matrix factorization model	Drug discovery, disease diagnosis and treatment	Novel method (GRGMF)

Citation	Year	Primary Objective	Biomedical Domain	Novel Method or Application
Daniel Neil et al.	2018	Develop interpretable GCNNs for inference on noisy knowledge graphs	Therapeutic target identification	Novel adaptation of GCNNs
Srivamshi Pittala et al.	2020	Identify disease genes using relation-weighted link prediction	Disease gene identification, drug target identification	Novel method
Simon Ott et al.	2022	Develop a software suite for predicting, explaining, and exploring links in large biomedical KGs	Hypothesis generation	Novel method (SAFRAN)
Finn Womack et al.	2019	Discover novel uses for known drugs by leveraging distributed biomedical knowledge sources	Drug repurposing	Adaptation of existing methods
Jaswanth K. Yella et al.	2022	Predict drug-target associations using multimodal graph neural networks	Drug discovery	Novel method (GraMDTA)
OUP accepted manuscripta	2021	Present a GNN based on link representation for predicting molecular associations	Biological process understanding	Novel method (LR-GNN)
Nansu Zong et al.	2017	Deep mine heterogeneous networks of biomedical linked data to predict novel drug-target associations	Drug discovery	Adaptation of DeepWalk
Changsung Moon et al.	2021	Develop a drug-disease-target embedding algorithm to inform drug repurposing hypotheses	Drug repurposing	Novel method

Citation	Year	Primary Objective	Biomedical Domain	Novel Method or Application
R. T. Sousa et al.	2023	Propose an explainable representation approach for relation prediction in KGs	PPI prediction, gene-disease association prediction	Novel method (SEEK)
Praveen Kumar et al.	2025	Integrate KGs and positive unlabeled learning for identifying disease-associated genes	Bioinformatics, disease genetics	Novel method (KG2ML)
R. Çelebi et al.	2018	Evaluate KGE approaches for DDI prediction using Linked Open Data	Drug design, drug safety	Adaptation of existing methods
Andrej Kastrin et al.	2014	Propose a methodology for link prediction of implicit relationships in the Semantic MEDLINE network	Literature-based discovery	Adaptation of existing methods
Khushnood Abbas et al.	2021	Apply network link prediction in drug discovery	Drug discovery, network biology	Adaptation of existing methods
D. Domingo-Fernández et al.	2021	Use causal reasoning over KGs for drug discovery leveraging transcriptomic signatures	Drug discovery, target discovery, drug repurposing	Novel method (RPath)
Ahmad F. Al Musawi et al.	2023	Review link prediction applications in network biology	Network biology	Review of existing methods
Clint Cuffy et al.	2025	Predict implicit concept embeddings for singular relationship discovery	Literature-based discovery, drug repurposing	Novel method

Citation	Year	Primary Objective	Biomedical Domain	Novel Method or Application
L. Eronen et al.	2012	Predict links between biological entities using network models of heterogeneous databases	Human genetics, PPI prediction, disease gene prioritization	Adaptation with novel components
D. Bang et al.	2023	Propose a "semantic multi-layer guilt-by-association" approach for drug repurposing	Drug repurposing	Novel adaptation of random walk (DREAMwalk)
Sachin Gavali et al.	2022	Predict novel kinase-substrate interactions using a KG representation learning approach	Kinase biology, molecular interactions	Novel adaptation (TripleWalk)
S. Paliwal et al.	2020	Predict therapeutic targets using tensor factorization on heterogeneous graphs	Drug discovery, target identification	Novel method (Rosalind)
Ana Beatriz Oliveira Villela Silva et al.	2021	Predict novel lncRNA-disease associations using graph convolutional auto-encoders	Disease biology, genomics	Novel application (PANDA)
Wen Tao et al.	2024	Bridge chemical structure and conceptual knowledge for compound-protein interaction prediction	Drug discovery	Novel framework (BEACON)
Yongkang Xiao et al.	2023	Repurpose drugs for Alzheimer's disease through link prediction on biomedical literature	Drug repurposing, neurology	Novel methodology
Yuriy Hulovatyy et al.	2014	Reveal missing parts of the interactome via link prediction	Proteomics, systems biology	Novel method

Citation	Year	Primary Objective	Biomedical Domain	Novel Method or Application
Chuanze Kang et al.	2021	Develop a GNN based on link representation for predicting molecular associations	Molecular biology, computational biology	Novel method (LR-GNN)
Ariam Rivas et al.	2023	Propose a neuro-symbolic system over knowledge graphs for link prediction	Lung cancer treatment effectiveness prediction	Novel hybrid system
Yushan Liu et al.	2020	Integrate logical rules into neural multi-hop reasoning for drug repurposing	Drug repurposing	Novel method
Ting Jia et al.	2022	Predict links on the KG of COVID-19 antiviral drugs using tensor decomposition	COVID-19 drug discovery	Adaptation with novel fusion strategy
Zhenxiang Gao et al.	2022a	Develop a KG-driven disease-gene prediction system using multi-relational GCNs	Disease-gene prediction, molecular mechanisms of disease	Not mentioned
Weichuan Wang et al.	2021	Propose a multiple scoring functions model for link prediction on antiviral drugs KG	COVID-19 research, drug discovery	Novel adaptation (MDistMult)
Brandon M. Malone et al.	2018	Use KG completion to predict polypharmacy side effects	Pharmacovigilance, drug safety	Adaptation of existing methods
Anna Breit et al.	2019	Introduce a resource and benchmarking framework for large-scale biomedical link prediction	Biomedical link prediction	Not mentioned

Citation	Year	Primary Objective	Biomedical Domain	Novel Method or Application
Daniel N. Sosa et al.	2019	Use a literature-based KGE method to identify drug repurposing opportunities in rare diseases	Drug repurposing, rare diseases	Novel in context
Galadriel Brière et al.	2025	Benchmark the impact of data leakage on KGE model performance for biomedical link prediction	Drug repurposing	Adaptation of existing methods
Julie Jiang et al.	2020	Learn graph representations of biochemical networks for enzymatic link prediction	Biological engineering, biological discovery	Novel method (ELP)
Susana Nunes et al.	2025	Reward explainability in drug repurposing with knowledge graphs	Drug repurposing, explainable AI	Novel method (REx)
Guillermo Palma et al.	2014	Predict drug-target interactions using semantic similarity and edge partitioning	Drug repurposing, drug safety surveillance	Novel method (semEP)
Cheng Chen et al.	2025	Predict links in multipartite graphs with application to drug repositioning studies	Drug repositioning	Novel method
A. Cattaneo et al.	2024	Investigate the role of graph topology in the performance of biomedical KGC models	Drug repurposing, drug-target identification	Adaptation of existing methods

Citation	Year	Primary Objective	Biomedical Domain	Novel Method or Application
Hajer Akid et al.	2024	Propose a graph-based machine learning model for weight prediction in PPI networks	Proteomics, network biology	Novel method
Joana Vilela et al.	2021	Use KGEs for personalized medicine applications	Personalized medicine, neurology (ASD)	Adaptation of existing methods
Jens Dörpinghaus et al.	2022	Propose a novel link prediction approach on clinical knowledge graphs using graph structures	Clinical data integration, medical diagnosis	Novel method
Soham Dasgupta et al.	2021	Develop an ADE prediction algorithm using noisy literature-derived knowledge graphs	Pharmacovigilance, adverse drug event discovery	Adaptation of existing methods
Chao Jiang et al.	2022	Develop a framework for deep denoising of raw biomedical knowledge graphs from COVID-19 literature	Data quality improvement, association prediction	Adaptation of existing methods
Shahin Ghasemi et al.	2023	Predict drug-disease associations based on end-to-end multi-layer heterogeneous graph convolutional encoders	Drug development, drug repositioning	Novel method (LHGCE)
Asan Agibetov et al.	2018	Propose a fast and scalable learning method for neuro-symbolic representations of biomedical knowledge	Biological link prediction	Adaptation with improvements

Citation	Year	Primary Objective	Biomedical Domain	Novel Method or Application
Gang Fu et al.	2016	Predict drug target interactions using meta-path-based semantic network analysis	Drug discovery, drug repositioning	Adaptation of existing methods
Yiding Lu et al.	2017	Predict links in drug-target interactions network using similarity indices	Drug repositioning	Novel application of existing methods
Tiyao Liu et al.	2025	Propose a novel approach for drug-disease association prediction using graph inference techniques	Pharmaceutical research and development	Novel method (TARMDNGI)
W. Djeddi et al.	2023	Propose a comprehensive graph-based approach for DTI prediction integrating KGE and ProtBert	Drug discovery	Novel method (DTIOG)
Yang-Han Wu et al.	2023	Propose a KGE model for profiling the interaction between transcription factors and target genes	Gene regulation network analysis	Novel method (KGE-TGI)
Sayantan Basu et al.	2020	Propose a pipeline for association analysis of COVID-19 through mining of biomedical corpora	COVID-19 research	Adaptation of existing methods
Özge Erten et al.	2023	Predict missing annotations in Gene Ontology with KGE and True Path Rule	Molecular biology, gene function prediction	Adaptation of existing methods

Citation	Year	Primary Objective	Biomedical Domain	Novel Method or Application
T. V. Ivanisenko et al.	2024	Propose a hybrid approach for knowledge extraction from scientific publications	Therapeutic target discovery, complex disorders research	Adaptation of existing methods
Edoardo Ramalli et al.	2021	Demystify drug repurposing domain comprehension with knowledge graph embedding	Drug repurposing	Adaptation of existing methods
Alexandre Renaux et al.	2023	Propose a knowledge graph approach to predict and interpret disease-causing gene interactions	Genetic disease research, oligogenic diseases	Novel application
Jeremy J. Yang et al.	2021	Develop a knowledge graph analytics platform for Parkinson's disease target illumination	Drug target hypothesis prioritization, complex diseases	Adaptation of statistical methods
Timo Sztyler et al.	n.d.	Propose a knowledge graph-based approach to predict novel relations	Disease-gene associations, polypharmacy side effects	Novel method (KBLRN)
Niclas Heilig et al.	2022	Refine diagnosis paths for medical diagnosis based on an augmented knowledge graph	Medical diagnosis	Adaptation of existing methods
Stanford CS224N et al.	2023	Use KGEs from biomedical language models to infer drug repurposing candidates for rare diseases	Drug repurposing, rare diseases	Adaptation of existing methods

Citation	Year	Primary Objective	Biomedical Domain	Novel Method or Application
G. Edwards et al.	2021	Explore explainable biomedical recommendations via reinforcement learning reasoning on KGs	Drug discovery, transparent AI in biology	Adaptation of existing methods
Saatviga Sudhahar et al.	2024	Propose an experimentally validated approach to automated biological evidence generation in drug discovery	Drug repositioning, explainable AI	Adaptation of existing methods
Yun Xiong et al.	2018	Predict disease-related associations by heterogeneous network embedding	Systems biology, complex diseases research	Novel method
Mustafa Coşkun et al.	2021	Use node similarity-based graph convolution for link prediction in biological networks	Network biology	Adaptation of GCNs (SiGraC)
Rahul Nadkarni et al.	2021	Study scientific language models for biomedical knowledge base completion	Drug design and repurposing	Adaptation with novel routing mechanism
Francesco Madeddu et al.	2025	Build a knowledge graph for biologically relevant learning tasks	Drug repurposing, PPI prediction, side-effect prediction	Adaptation of existing methods
Mariah Salcedo et al.	2023	Predict protein and pathway associations for understudied dark kinases using KG embedding	Kinase function prediction, bioinformatics	Novel method (RegPattern2Vec)
Yitong Lin et al.	2025	Propose a framework for synergistic semantic and structural learning in biomedical KGs	Drug discovery, disease understanding	Novel framework (BioGraphFusion)

Citation	Year	Primary Objective	Biomedical Domain	Novel Method or Application
Haitao Fu et al.	2021	Propose a multi-view graph convolutional network for predicting links in biomedical bipartite networks	Disease diagnosis and treatment	Novel method (MVGCN)
Armando Vieira et al.	2016	A GCN based KG visualization tool	DTI Prediction	Novel Application
Yue Hong et al.	2022	Propose a link-aware graph attention network for DDI prediction	Pharmacology, clinical applications	Novel method (LaGAT)
Daniel Rivas-Barragan et al.	2022	Use ensembles of KGE models to improve predictions for drug discovery	Drug discovery	Adaptation of existing methods
Disha Purohit et al.	2025	Propose a neuro-symbolic system for inductive learning over medical KGs	Medical knowledge discovery, lung cancer care	Adaptation of KGE methods
U. M. Singh-Blom et al.	2013a	Predict and validate gene-disease associations using methods inspired by social network analyses	Disease biology, genomics	Adaptation of existing methods
Mustafa Coşkun et al.	2020	Use proximity measures as graph convolution matrices for link prediction in biological networks	Disease gene prioritization, drug-disease associations	Adaptation with novel integration
R. Mythili et al.	2023	Predict links of biomedical data using a Hetrosim GCN-GAT model	DDI prediction, drug effectiveness	Novel method

Computational Methods and Algorithmic Approaches

The computational methods employed for link prediction in biomedical knowledge graphs are diverse, spanning several major categories. Knowledge graph embedding (KGE) methods are widely used, including translational

distance models like TransE, TransH, TransD, and semantic matching models such as RESCAL, DistMult, ComplEx, and RotatE. These methods aim to represent entities and relations as low-dimensional vectors to infer missing links. Key innovations in this area include the development of novel models like TriModel, which extends DistMult and ComplEx with a generative approach based on tensor factorization, and SimTransE, which incorporates domain similarity values into the TransE framework. Other works focus on integrating additional information, such as combining KGE with text embeddings from large language models (LLMs), or using multi-part embedding vectors with tensor decomposition.

Graph Neural Networks (GNNs) represent another major category, leveraging graph structure to learn node representations. This includes Graph Convolutional Networks (GCNs) and specialized architectures. For example, BioPathNet is a novel GNN framework based on the Neural Bellman-Ford Network that uses path-based reasoning and a background regulatory graph for enhanced message passing. Other GNN-based innovations include DenoisedLP, which introduces a denoising module to handle unreliable interactions, and LR-GNN, which focuses on learning representations of links rather than just nodes. KnowDDI enhances drug representations by adaptively leveraging neighborhood information and learning interpretable subgraphs.

Ensemble and hybrid approaches are also common. Several studies combine predictions from multiple KGE models or integrate KGE methods with path-based reasoning approaches to increase confidence in predictions. Neuro-symbolic methods merge neural networks with logical rules to improve reasoning capabilities and interpretability, as seen in the PoLo method, which combines reinforcement learning-guided walks with logical rules integrated via a novel reward function. Other unique approaches include using diffusion-based methods on multimodal networks, applying enrichment analysis to infer edges, and employing generative adversarial networks to denoise raw knowledge graphs. These methods are often designed to handle the specific challenges of biomedical KGs, such as heterogeneity, which is addressed through multi-relational GNNs, path-based analyses, and the integration of diverse data modalities.

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Yue Hu et al.	BioPathNet	Graph Neural Network	Path-based reasoning, background regulatory graph, stringent negative sampling	BRG used for enhanced message passing across different relation types	Novel
Yue Hu et al.	BioPathNet	Graph Neural Network	Path-based reasoning based on NBFNet, use of background regulatory graph (BRG)	Uses BRG to integrate regulatory information into message passing	Novel Adaptation
Joana Vilela et al.	ComplEx, DistMult, TransE	Knowledge Graph Embedding	Application of KGE to a custom KG for gene-disease prediction, use of l2-regularization	Models are inherently capable of handling multi-relational data	Adaptation

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Haji Gul et al.	MuCoS	Knowledge Graph Embedding	Prioritizes high-density neighbors and integrates BERT-derived contextual embeddings	Integrates structural patterns with textual context from neighbors	Novel Adaptation
Tengfei Ma et al.	DenoisedLP	Knowledge Graph Embedding / GNN	Denoising module (SRL), smooth semantic preservation, mutual information maximization	Smooths semantic information by blurring conflict relations	Novel Framework
Meng Wang et al.	Predicting Rich DDI	Knowledge Graph Embedding	Joint translation-based embedding integrating KGs and biomedical text; predicts multiple DDI labels	Integrates structured KG data and unstructured text into a common space	Novel
Yongkang Xiao et al.	FuseLinker	GNN / KGE	Fuses pre-trained text embeddings (from LLMs) and domain knowledge embeddings (from ontologies)	Fuses graph structure, text, and domain knowledge information	Novel
Yahui Long et al.	PT-GNN	Graph Neural Network	Pre-training framework using CNN and GCN to learn and refine node features from sequence/structure data	Integrates features from different data sources (sequence, structure, network)	Novel Framework

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Zhenxiang Gao et al.	KG-Predict	Knowledge Graph Embedding	Aggregation of heterogeneous topological and semantic information from a custom KG	Learns low-dimensional representations from heterogeneous data sources	Novel
V. Nováček et al.	Tensor Decomposition	Knowledge Graph Embedding	Uses multi-part embedding vectors and tensor decomposition for link prediction	Not mentioned	Novel
Chih-Hsu Lin et al.	GID, AptRank, RW	Diffusion-based	Application of diffusion methods on a multimodal network with multiple edge types	Models information flow across different association types in a multimodal network	Adaptation
Yaqing Wang et al.	KnowDDI	Graph Neural Network	Adaptive leveraging of neighborhood information, learning of interpretable knowledge subgraphs	Merges DDI graph with an external KG to create a combined network	Novel Adaptation
M. Chang et al.	CMEI	LLM / KGE	Integrates embeddings from different modalities (LLM and KGE) using a multi-head attention mechanism	Fuses embeddings from textual (LLM) and structural (KGE) modalities	Novel
M. O. Faruk et al.	Bilinear model	GNN / KGE	Implemented in PyTorch Geometric on a large-scale pharmacogenomic KG	Integrates and harmonizes data from multiple databases (PharmGKB, ClinVar, etc.)	Adaptation

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Nansu Zong et al.	Ensemble Learning	Knowledge Graph Embedding	Uses Node2Vec for feature learning and integrates inference-based and classification models	Constructs a heterogeneous network incorporating 7 types of biomedical entities	Adaptation
Lizzy Farrugia et al.	ComplEx+LSTM, GNN Auto-encoder	KGE / GNN	Combination of ComplEx with an LSTM network; a GNN-based graph auto-encoder	Integrates multiple drug features from various repositories into a single KG	Adaptation
F. Aisopos et al.	BLGPA, TransE, etc.	Path-based / KGE	Explainable path analysis technique and use of a decision tree for explainability	Handles relations from literature, such as "MENTIONED_IN" and "HAS_MESH" in the KG	Adaptation
Catarina Canastra et al.	TransD, HolE, etc.	Knowledge Graph Embedding	Novel framework for comparing link prediction vs. node-pair classification tasks	Integrates various knowledge graphs and ontologies (GO, HPO, DO)	Adaptation in a Novel Framework
Yang Zhao et al.	TransE, ComplEx, etc.	Knowledge Graph Embedding	Abstraction of a Community KG from a larger KG; extension approach to link the two graphs	Not mentioned	Adaptation
Kevin McCoy et al.	TransE, RotatE, ComplEx	Knowledge Graph Embedding	Use of DGL-KE library for scalable training on a large literature-derived KG	Uses an existing KG (SemNet) with 132 entity types and 61 relation types	Adaptation

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Yushan Liu et al.	PoLo	Neuro-symbolic / Reinforcement Learning	Combines policy-guided walks with logical rules via a novel reward function	Integrates domain knowledge via logical rules	Novel Integration
Vignesh Prabhakar et al.	Ensemble Learning	Ensemble / KGE	Divides full KG into smaller subgraphs and aggregates link predictions	Manages heterogeneity by training models on smaller, more localized subgraphs	Adaptation
Roger Tu et al.	Consilience Ensemble	Ensemble (KGE + Path-based)	Combines reasoning paths and predictions from KGE and path-based methods	Integrates different reasoning paradigms (vector-based and path-based)	Adaptation
Sameh K. Mohamed et al.	TriModel	Knowledge Graph Embedding	Extends Dist-Mult/ComplEx using tensor factorization; generative approach for efficient representations	Encodes multiple association types within the model	Novel
Gamal K. O. Crichton et al.	Neural Link Predictor	Knowledge Graph Embedding	Uses various KGE methods (DeepWalk, LINE, etc.) as inputs to a neural classifier	Evaluates performance on graphs with varying sizes and connectivity	Adaptation
Jinjiang Guo et al.	GPLP	Graph Neural Network	Uses 1-hop subgraphs extracted from the interaction matrix to predict links	Evaluated on heterogeneous biomedical networks (DTI, CPI, CVI)	Novel

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Budhaditya Saha et al.	Neighborhood-based embedding	Knowledge Graph Embedding	Novel application to simultaneously predict both noisy and missing links	Not mentioned	Novel Application
Sameh K. Mohamed et al.	Customized ComplEx	Knowledge Graph Embedding	A customized version of the ComplEx model	Applied to a KG with drugs, proteins, diseases, and pathways	Novel Adaptation
Nansu Zong et al.	Ensemble Method	Knowledge Graph Embedding / Ensemble	Modularization of feature selection and association prediction	Benchmarked on a comprehensive heterogeneous biomedical network	Adaptation
Tien Dang et al.	Multimodal Approach	LM / GCL / KGE	Unifies embeddings from LMs with Graph Contrastive Learning and KGE models	Integrates embeddings from different modalities using a Fusion Module	Novel
Mona Alshahrani et al.	TransE, R-GCN, etc.	Knowledge Graph Embedding	Comparative assessment of various KGE methods on biomedical graphs	Uses RDF as a graph data model to handle different relation types	Adaptation
Xiaomin Liang et al.	Cascade Learning Framework	KGE / Graph Embedding	Integrates semantic features from KGE and graph features from graph embedding	Uses a meta-path algorithm to detect path relations in the network	Adaptation with Novel Framework
U. M. Singh-Blom et al.	Katz measure, Catapult	Graph-based / Machine Learning	Catapult learns feature weights from walks in a heterogeneous network	Catapult operates on a heterogeneous gene-trait network	Adaptations

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Wenhui Wang et al.	HGBI	Graph-based Inference	Iterative updates using all paths in the network; incorporates similarity information	Specifically designed for heterogeneous graphs with different node types (drugs, targets)	Novel Adaptation
Gokhan Bakal et al.	Logistic Regression, Decision Tree	Graph Pattern Feature Extraction	Uses frequencies of semantic graph patterns connecting entities as features	Extracts semantic patterns from a heterogeneous KG (SemMedDB)	Novel Application
A. Daowd et al.	FocusE-TransE	Knowledge Graph Completion	Application of KGC to predict missing relations between head and tail entities	Applied to a cancer-focused KG with drug-gene-disease paths	Adaptation
D. Bean et al.	Simple Enrichment Test	Machine Learning	Uses Fisher's exact test to identify enriched properties for binary classification	Integrates drugs, protein targets, indications, and adverse reactions in a single graph	Adaptation
Mona Alshahrani et al.	Word2Vec, TransE	KGE / Text Analysis	Integration of KGs with text data using named entity recognition and normalization	Combines structured data from KGs and unstructured text from literature	Adaptation with Innovative Integration
P. Sevon et al.	Two-terminal network reliability	Graph-based Reliability	Uses edge weights based on reliability, relevance, and rarity for link goodness	Represents data from heterogeneous databases as a graph with multiple edge types	Novel Application
Meghamala Sinha et al.	WeightedKgBlend	Ensemble (KGE + Path-based)	Dynamically assigns weights to models based on individual performance	Not mentioned	Novel

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
R. Çelebi et al.	RDF2Vec, TransE, TransD	Knowledge Graph Embedding	Use of RDF2Vec with Skip-Gram for DDI prediction	Evaluated on integrated KGs from DrugBank, PharmGKB, and KEGG	Adaptation
Ankit Patel et al.	LINE, HOPE, SDNE, GraphSAGE	Knowledge Graph Embedding	Combination of embedding techniques with various machine learning classifiers	Analyzes a heterogeneous KG of concepts and papers from literature	Adaptation
Jake Lever et al.	SVD	Matrix Factorization	Uses SVD to combine data from across the entire knowledge graph	Not mentioned	Adaptation
Awais Bajwa et al.	SimTransE	Knowledge Graph Embedding	Incorporates domain similarity values between drugs and targets into the TransE model	Computes and integrates drug-drug and target-target similarities	Adaptation with Novel Modification
Olawumi Olasunkanmi et al.	EDGAR	Enrichment-driven method	Leverages enrichment analysis to mine entity-local rules for explainable edge inference	Applied to a large, heterogeneous biomedical KG (ROBOKOP)	Novel
Alejandro Flores et al.	esDSG	Densest Subgraph Algorithm	Incorporates semantic similarity of nodes into a densest subgraph algorithm	Exploits semantic knowledge in conjunction with topological information	Adaptation

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Susana Nunes et al.	TransE, HAKE, etc.	Knowledge Graph Embedding	Uses multiple interconnected ontologies linked by logical definitions and mappings	Integrates multiple domains (phenotypes, gene functions) via ontology mappings	Adaptation
Aryo Pradipta Gema et al.	ComplEx, ConvE, AnyBURL	KGE / Rule Learning	Use of pretrained models and a transfer learning paradigm for downstream tasks	Evaluated on BioKG, a comprehensive KG covering multiple biomedical entities	Adaptation
OUP accepted manuscript	SAFRAN	Rule-based	Novel rule-based link prediction engine	Applied to large biomedical knowledge graphs	Novel
V. Nováček et al.	ComplEx	Knowledge Graph Embedding	Application of ComplEx leveraging latent features and long-range relationships for kinase-substrate prediction	Represents phosphorylation networks as knowledge graphs	Adaptation
K. Bougiatiotis et al.	DDI-BLKG	Knowledge Graph Embedding	Uses semantic relations in paths as features, with one-hot encoding of relations	Integrates hierarchical relations from multiple biomedical ontologies	Novel Adaptation
Mona Alshahrani et al.	Modified DeepWalk	Knowledge Graph Embedding	Integrates symbolic logic and automated reasoning by incorporating edge labels into the walk	Deductively closes the KG to infer new triples before embedding	Adaptation with Novel Integration

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Binon Teji et al.	Graph Embedding	Knowledge Graph Embedding	Evaluation of multiple embedding techniques combined with various similarity functions for reconstruction	Evaluates performance on both homogeneous and heterogeneous biological networks	Adaptation
Ozge Gurbuz et al.	Shortest path, RDF2Vec, etc.	KGE / Machine Learning	Use of tissue expression information in shortest path methods; application of Random Forest with embeddings	Combines literature-based interactions with gene expression data in tissues	Adaptation
Xiang Yue et al.	Various	Graph Embedding	Systematic comparison of matrix factorization, random walk, and neural network-based methods	Applied to biomedical networks with different entity and relation types	Adaptations
Daniel N. Sosa et al.	TransE, RotatE, etc.	Knowledge Graph Embedding	Downsampling high-degree triples to mitigate topological bias	Evaluated on graphs from structured databases (Hetionet) and literature (GNBR)	Adaptation
Daniel Daza et al.	BioBLP	Knowledge Graph Embedding	Uses modality-specific encoders for different attribute data types; efficient pretraining strategy	Supports entities with heterogeneous and missing data modalities	Novel

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Francesco Gualdi et al.	DLemb, BioKG2Vec	Knowledge Graph Embedding	Novel KGE algorithms developed for this study	Integrates heterogeneous biomedical data for diseases with limited information	Novel
Ziyan Zhang et al.	GRGMF	Matrix Factorization / KGE	Incorporates neighborhood information and graph regularization terms from external databases	Integrates affinity information from external databases via regularization	Novel
Daniel Neil et al.	GCNNs with Attention	Graph Neural Network	Introduction of a regularized attention mechanism to GCNNs to handle noise	Addresses noise, a pervasive issue in multi-source biomedical KGs	Novel Adaptation
Srivamshi Pittala et al.	Not specified	Knowledge Graph Embedding	Not specified	Studies the effects of various relation types on performance	Novel
Simon Ott et al.	SAFRAN	Rule-based an	A novel, rule-based link prediction engine that provides explanations	Designed for large biomedical knowledge graphs	Novel
Finn Womack et al.	node2vec + Random Forest	Knowledge Graph Embedding	Uses node2vec embeddings as features for a random forest model	Integrates 20 repositories with 11 distinct bioentity types	Adaptation
Jaswanth K. Yella et al.	GraMDTA	Multimodal GNN	Fuses structural (molecular graphs) and KG-based representations using multi-head attention	Integrates two data modalities (structure and interactions) for DTA prediction	Novel

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
OUP accepted manuscripta	LR-GNN	Graph Neural Network	Uses a propagation rule to construct link representations and a layer-wise fusing rule	Applied to various biomedical networks (lncRNA-disease, DDI, etc.)	Novel
Nansu Zong et al.	DeepWalk	Knowledge Graph Embedding	Uses DeepWalk to calculate similarities within a heterogeneous Linked Tripartite Network	Constructs and analyzes a tripartite network of drugs, targets, and diseases	Adaptation
Changsung Moon et al.	New embedding algorithm	Knowledge Graph Embedding	Represents predicates as translation vectors between subject and object entity vectors	Applied to a drug-disease-target network	Novel
R. T. Sousa et al.	SEEK	Knowledge Graph Embedding	Generates explainable representations based on shared semantic aspects (subgraphs)	Identifies relevant shared semantic aspects between entities of different types	Novel
Praveen Kumar et al.	KG2ML	PU Learning / Machine Learning	Integrates Positive Unlabeled (PU) learning with path-based feature extraction from a KG	Extracts path-based features from a heterogeneous KG (DDKG)	Novel
R. Çelebi et al.	RDF2Vec, TransE, TransD	Knowledge Graph Embedding	Use of RDF2Vec with uniform weighting and Skip-Gram architecture	Applied to a KG built from Linked Open Data (Bio2RDF)	Adaptation

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Andrej Kastrin et al.	Common neighbors, etc.	Link Prediction	Application of traditional similarity measures to a large semantic literature network	The Semantic MEDLINE network contains various biomedical concepts and relationships	Adaptation
Khushnood Abbas et al.	Prone, ACT, etc.	Network-based Machine Learning	Converts link prediction to a binary classification problem	Applied to various biomedical networks (DGA, DDA, DTI, DDI)	Adaptation
D. Domingo-Fernández et al.	RPath	Causal Reasoning	Reasons over causal paths in a multimodal KG guided by transcriptomic signatures	Leverages gene expression signatures to guide reasoning over a multimodal KG	Novel
Ahmad F. Al Musawi et al.	Review	Review	Systematic dissection of local, centrality, and embedding-based LP approaches	Reviews applications across heterogeneous networks (diseases, genes, drugs, etc.)	Review
Clint Cuffy et al.	Interconnecting Vertex Embedding	Knowledge Graph Embedding	Re-frames KGC as predicting interconnecting vertex embeddings using a similarity learning objective	Explores three different ways to represent edge connections (average, concat, Hadamard)	Novel
L. Eronen et al.	Random walk, Clustering	Graph-based Proximity	Weights edges based on relevance, informativeness, reliability; novel clustering method	Integrates multiple databases into a graph with multiple edge types	Adaptation with Novel Components

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
D. Bang et al.	DREAMwalk	Knowledge Graph Embedding	Uses a semantic information-guided random walk to integrate molecular and semantic levels	Extends guilt-by-association to multiple layers (drug-gene-disease)	Novel Adaptation
Sachin Gavali et al.	TripleWalk	Knowledge Graph Embedding	Modified random walk approach incorporating the triple structure (head, relation, tail)	Considers directionality of interactions in the phosphoproteomic KG	Novel Adaptation
S. Paliwal et al.	Rosalind	Tensor Factorization / KGE	Relational inference via tensor factorization on a heterogeneous knowledge graph	Explicitly represents multiple entity relations from heterogeneous data sources	Novel Combination
Ana Beatriz Oliveira Villela Silva et al.	PANDA	Graph Neural Network	Treats prediction as a link prediction problem using a Graph Auto-encoder	Constructs a heterogeneous graph including associations, expression levels, and symptoms	Novel Application
Wen Tao et al.	BEACON	Self-supervised Learning / KGE	Dual contrastive and dual predictive learning to bridge chemical structure and conceptual knowledge	Addresses knowledge-missing entities by predicting their knowledge representations	Novel Framework
Yongkang Xiao et al.	R-GCN, CompGCN, etc.	KGE / GNN	Comparison of multiple KGE and GCN models to find the best performer for the task	Applied to a KG extended with semantic triples from SemMedDB	Novel Methodology

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Yuriy Hulovatyy et al.	Graphlet-based measures	Link Prediction	Uses novel sensitive LP measures based on shared extended neighborhoods and graphlets	Focuses on protein interaction networks (PINs), a type of homogeneous graph	Novel
Chuanze Kang et al.	LR-GNN	Graph Neural Network	Designs a propagation rule to construct link representations from node embeddings	Not mentioned	Novel
Ariam Rivas et al.	Hybrid neuro-symbolic system	KGE / Symbolic Reasoning	Augments entity context with new RDF triples deduced by a symbolic system before embedding	Deduces new relationships within ego networks to handle sparsity and incompleteness	Novel
Yushan Liu et al.	Neural multi-hop reasoning	Reinforcement Learning / GNN	Integration of logical rules into the neural multi-hop reasoning process	Handles long-range dependencies common in biomedical data via logical rules	Novel
Ting Jia et al.	Fusion Strategy	Ensemble Learning	Integrates predictions from two tensor decomposition models (CP-N3 and ComplEx-N3)	Analyzes performance differences across various relationship types	Adaptation with Novel Fusion
Zhenxiang Gao et al.	GenePredict-KG	Knowledge Graph Embedding	Models semantic relations from various genotypic and phenotypic databases	Constructs a KG with 14 types of phenotypic and genotypic relations and 7 entity types	Not mentioned

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Weichuan Wang et al.	MDistMult	Knowledge Graph Embedding	Leverages multiple DistMult scoring functions to extract more features and handle asymmetry	Designed for KGs with potentially asymmetric relations, common in biology	Novel Adaptation
Brandon M. Malone et al.	DistMult, KBlrn	Knowledge Graph Embedding	Incorporation of interpretable and rule-based features into the KGE model	Models multi-relational data including protein-protein and drug-protein interactions	Adaptation
Anna Breit et al. Daniel N. Sosa et al.	Not mentioned Uncertain KGE	Not mentioned Knowledge Graph Embedding	Not mentioned Explicitly models uncertainty associated with literature-derived relationships	Not mentioned Handles noisy relationships derived from text mining PubMed abstracts	Not mentioned Novel in context
Galadriel Brière et al.	TransE, ComplEx, etc.	Knowledge Graph Embedding	Systematic evaluation of data leakage's impact on KGE model performance	Evaluation of performance on cold-start vs. random splits to mimic real-world scenarios	Adaptation
Julie Jiang et al.	ELP	Knowledge Graph Embedding	Uses graph embedding to capture molecular attributes, enzymatic attributes, and graph connectivity	Models enzymatic reactions from KEGG as a graph	Novel

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Susana Nunes et al.	REx	Reinforcement Learning	Uses reward and policy mechanisms to guide an RL agent in finding explanatory paths	Enriches explanatory paths with domain-specific ontologies (NCIT, ChEBI)	Novel
Guillermo Palma et al.	semEP	Edge Partitioning / Data Mining	Combines semantic similarity with edge partitioning to create clusters of interactions	Partitions edges of a heterogeneous graph including drug-target and similarity edges	Novel
Cheng Chen et al.	Not mentioned	Graph Theoretical	An innovative technique to impute potential edges in an arbitrary multipartite graph	Specifically designed for multipartite graphs with three or more partite sets	Novel
A. Cattaneo et al.	TransE, RotatE, etc.	Knowledge Graph Embedding	Use of Graphcore IPU's and BESS framework for efficient training	Investigates how topological properties of different biomedical KGs impact model performance	Adaptation
Hajer Akid et al.	Graph-based regression tree	Machine Learning	Integrates topological measures with aggregation-based representations for weight prediction	Applied to a weighted PPI network where edge weights denote confidence scores	Novel
Joana Vilela et al.	Not specified	Knowledge Graph Embedding	Not mentioned	Integrates data from Ensembl, DisGeNET, and Gene Ontology	Adaptation

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Jens Dörpinghaus et al.	Graph structures + CRFs	Not specified	Novel approach utilizing graph structures and Conditional Random Fields	Handles clinical imaging data in a knowledge graph representation	Novel
Soham Dasgupta et al.	Weighted DeepWalk, Weighted TransE	Knowledge Graph Embedding	Incorporates confidence scores from NLP extraction into the embedding process	Explicitly models noise and inaccuracies in a literature-derived KG	Adaptation
Chao Jiang et al.	NetGAN, CELL	Generative Deep Neural Networks	Leverages unlabeled link information to denoise biomedical knowledge graphs	Applied to a raw KG built from co-occurring entities in COVID-19 literature	Adaptation
Shahin Ghasemi et al.	LHGCE	Graph Neural Network	End-to-end multi-layer heterogeneous graph convolutional encoder	Creates and combines multiple networks to build a heterogeneous knowledge graph	Novel
Asan Agibetov et al.	Log-linear neural embeddings	Knowledge Graph Embedding	Fast training and economical embedding dimensions; encodes directionality of asymmetric relations	Naturally encodes directionality by controlling the order of embedding concatenation	Adaptation with Improvements
Gang Fu et al.	Random Forest, SVM	Ensemble Learning	Use of meta-path topological features and Random Walk (RW) normalization	Employs meta-paths to navigate and featurize the heterogeneous semantic network	Adaptation

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Yiding Lu et al.	Similarity Indices	Link Prediction	Adaptation of similarity indices (Common Neighbours, etc.) for bipartite DTI networks	Specifically designed for bipartite networks where nodes belong to two distinct sets	Novel Application
Tiyao Liu et al.	TARMDNGI	Matrix Completion / Graph Inference	Truncated arctangent rank minimization; double-strategy neighborhood constrained inference	Employs dual strategies to process complex structures in heterogeneous graphs	Novel
W. Djeddi et al.	DTIOG	Knowledge Graph Embedding	Uses ProtBERT for target-target similarity and extracts local representations from SMILES/sequences	Builds and utilizes a weighted heterogeneous graph of drugs, targets, etc.	Novel
Yang-Han Wu et al.	KGE-TGI	Knowledge Graph Embedding	Multi-subgraph convolution network trained with a multi-task objective (link prediction + type classification)	Uses specific subgraphs to handle different relation types in the heterogeneous graph	Novel
Sayantan Basu et al.	TransD	Knowledge Graph Embedding	Benchmarking of different KGE techniques, with TransD found to be the best performing	Creates a KG from unstructured biomedical corpora (CORD-19)	Adaptation
Özge Erten et al.	KGE with TPR	Knowledge Graph Embedding	Application of the True Path Rule (TPR) during the training stage of KGE	Applied to the Gene Ontology knowledge base	Adaptation

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
T. V. Ivanisenko et al.	Hybrid GNN-LLM	GNN / KGE	Integration of GNNs with LLMs for validation and explanation of predicted edges	Extends biomedical KGs by analyzing literature	Adaptation
Edoardo Ramalli et al.	TransE	Knowledge Graph Embedding	Application to a KG designed for drug repurposing to improve predictions while saving resources	Focuses on how the graph's structure and entity types influence predictions	Adaptation
Alexandre Renaux et al.	Decision Set Classifier	Association Rule Mining	Trains an interpretable decision set model on heterogeneous paths connecting gene pairs	Integrates curated clinical cases with biomedical networks and ontologies	Novel Application
Jeremy J. Yang et al.	KG Analytics Platform	Statistical Aggregation	Uses Stouffer's function for z-score meta-analysis to score evidence paths	Integrates diverse datasets from LINCS (omics) and IDG (curated knowledge)	Adaptation of Statistical Methods
Timo Sztyler et al.	KBLRN	Graph Neural Network	Complements a neural relational approach with path features in a mixture-of-experts model	Not mentioned	Novel
Niclas Heilig et al.	RDF2vec	Knowledge Graph Embedding	Uses RDF2vec to create latent representations for refining diagnosis paths in a medical KG	Enriches a medical KG with patient data from MIMIC-IV	Adaptation

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Stanford CS224N et al.	KGE with LMs	Knowledge Graph Embedding	Leverages language models and biomedical corpora to create KG embeddings	Uses textual components from the GNBR network in the embedding process	Adaptation
G. Edwards et al.	Neurosymbolic Reasoning	Reinforcement Learning	Systematic application of multi-hop reasoning on KGs to multiple biomedical datasets	Not mentioned	Adaptation
Saatviga Sudhahar et al.	AnyBURL with filtering	Symbolic Reasoning	Combines a symbolic KGC model with an automatic filtering approach for relevant rules	Applied to a proprietary KG (Healx KG) with multiple node and edge types	Adaptation
Yun Xiong et al.	Network Embedding	Knowledge Graph Embedding	Learns a low-dimensional vector space that preserves relationships from multiple databases	Integrates several public databases to construct a heterogeneous network	Novel
Mustafa Coşkun et al.	SiGraC	Graph Neural Network	Uses node similarity-based convolution matrices in a single-layer GCN architecture	Applied to various biomedical networks including DDA, DDI, and PPI	Adaptation
Rahul Nadkarni et al.	Router Method	KGE / LM Integration	Integrates KGE and LM models using a router that learns to assign examples to either model type	Enriches KGs with textual entity descriptions to leverage LMs	Adaptation with Novel Routing

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Francesco Madeddu et al.	R-GCN, CompGCN	Graph Neural Network	Introduction of node-type-specific linear transformations to support heterogeneous nodes	Enriches nodes with features like molecular fingerprints and GO terms	Adaptation
Mariah Salcedo et al.	RegPattern2Vec	Knowledge Graph Embedding	Employs regular pattern constrained random walks to sample diverse node context flexibly	Learns representations from a kinase-centric KG with 11 node types and 13 association types	Novel
Yitong Lin et al.	BioGraphFusion	KGE / GNN	Synergistic semantic (tensor decomposition) and structural (LSTM-driven GNN) learning	Employs query-guided subgraph construction to handle large, complex KGs	Novel Framework
Haitao Fu et al.	MVGCN	Graph Neural Network	Constructs a multi-view heterogeneous network and uses a neighborhood information aggregation layer	Combines similarity networks with a biomedical bipartite network into multiple views	Novel
Armando Vieira et al.	Compositional model	GCN / KGE	GAN for visualization of KG	Not mentioned	Novel Application
Yue Hong et al.	LaGAT	Graph Neural Network	Generates different attention pathways for drug entities based on different drug pair links	Considers semantic diversity of entity nodes in different drug pair links	Novel

Study	Method Name	Method Category	Key Innovations	Handling of Heterogeneity	Novelty
Daniel Rivas-Barragan et al.	Ensemble Learning	Ensemble	Application of ensemble methods to aggregate predictions from multiple KGE models	Applied to biomedical KGs designed for drug discovery	Adaptation
Disha Purohit et al.	Neuro-symbolic system	KGE / Symbolic Learning	Combines symbolic learning from ontologies, inductive learning via KGE, and semantic constraint validation	Incorporates semantic constraints from medical ontologies and clinical guidelines	Adaptation with Unique Integration
U. M. Singh-Blom et al.	Katz measure, CATAPULT	Graph-based / Machine Learning	CATAPULT uses a biased SVM to learn weights for walks in a heterogeneous network	CATAPULT derives features from a heterogeneous network of genes and traits	Adaptation
Mustafa Coşkun et al.	GCN with DGI	Graph Neural Network	Integration of node similarity measures as convolution matrices within the GCN encoder	Evaluated on different types of biomedical networks (DDI, DDA, PPI)	Adaptation with Novel Integration
R. Mythili et al.	HetrosimGCN-GAT	Graph Neural Network	GCN for embedding representation and GAT for weighting neighborhood node contributions	Builds a heterogeneous network from drug-drug, protein-protein, and drug-protein matrices	Novel

Knowledge Graph Construction and Data Sources

The knowledge graphs (KGs) utilized in the reviewed studies are either constructed de novo from multiple biomedical data sources or are adaptations of existing, publicly available KGs. Custom-built KGs often integrate data from a wide array of databases such as DrugBank, KEGG, PubMed/MEDLINE, STRING, DisGeNET, and Gene Ontology (GO). For instance, one study constructed a KG with over 93,000 nodes and 1.7 million edges from DisGeNET and GO to represent genes, diseases, and other biological entities. Another created PGx-KG, a large-scale graph with over 3.7 million nodes and 9.6 million edges, by integrating PharmGKB, ClinVar, SIDER, and Reactome.

Commonly used existing KGs include Hetionet, the Drug Repurposing Knowledge Graph (DRKG), and BioKG. These graphs are typically large and heterogeneous, containing millions of nodes and edges representing entities like drugs, genes, proteins, diseases, pathways, and side effects. The number of relation types can be extensive, with DRKG containing 107 relations and one custom graph including 59 types.

Data preprocessing is a critical step. It often involves mapping and normalizing identifiers from different sources to a consistent standard, such as UMLS, Ensembl, or PubChem identifiers. Some studies perform extensive cleaning, such as de-duplication, checking for symmetric and transitive consistency, and enforcing chronological data splits to prevent temporal leakage. Graph modification steps may include removing low-degree nodes to reduce noise, removing specific edge types to prevent information leakage during evaluation, or enriching the graph with additional features like molecular fingerprints or textual descriptions.

Study	Data Source(s)	# Nodes	# Edges	# Relation Types	Entity Types	Custom or Existing KG
Yue Hu et al.	KEGG, PrimeKG, SynLethDB, LncTarD	129,375 (PrimeKG), 30,885 (BRG)	8,100,498 (PrimeKG), 1,884,146 (BRG)	Not mentioned	Genes, pathways, drugs, diseases, lncRNAs	Custom KGs constructed
Joana Vilela et al.	DisGeNET, GO	93,657	1,705,585	59	Genes, diseases, molecular functions, cellular components, biological processes	Custom
Haji Gul et al.	KEGG50k	Not mentioned	63,080 triples	9	Drugs, genes, pathways, diseases	Existing
Tengfei Ma et al.	DRKG	97,238	5,874,261 triples	107	Drugs, genes, diseases	Existing
Meng Wang et al.	Bio2RDF, DrugBank	313,839	Not mentioned	1053 labels	Drugs, targets, pathways, phenotypes	Custom

Study	Data Source(s)	# Nodes	# Edges	# Relation Types	Entity Types	Custom or Existing KG
M. Faruk et al.	PharmGKB, ClinVar, SIDER, Reactome	3,744,727	9,645,367	6 families	Genes, variants, drugs, diseases, ADRs	Custom
Nansu Zong et al.	12 repositories, DrugBank	20,119	194,296	14	Drugs, targets, diseases, side effects, pathways	Custom
Lizzy Farrugia et al.	DrugBank, KEGG, PharmGKB, SIDER	18,322,022	180,800,521 triples	Not mentioned	Drugs, diseases, side effects	Custom
F. Aisopos et al.	PubMed, various online sources	Not mentioned	>670,000 pairs	Not mentioned	Drugs, genes	Custom
Kevin McCoy et al.	SemNet (PubMed), CORD-19	Not mentioned	22,000,000 non-generic triples	61	132 ontological types of biomedical concepts	Existing, augmented
Yushan Liu et al.	Hetionet (29 databases)	Not mentioned	775 "treats" links	Not mentioned	Compounds, diseases	Existing
Vignesh Prabhakar et al.	DRKG, iBKH, Hetionet, BioKG	Not specified	~2M to ~50M triples	Not specified	Drugs, diseases, genes	Existing
Nansu Zong et al.	DrugBank, Diseasome	10,413	22,000	Not specified	Drugs, targets, diseases	Custom
and many more...						

Biomedical Applications and Prediction Tasks

The primary application of link prediction in the reviewed literature is to uncover novel associations with therapeutic and biological significance. A dominant theme is drug repurposing, where studies predict new drug-disease interactions to find new uses for existing drugs. This task is often framed as predicting "TREATS" or similar relationships between drug and disease entities in the knowledge graph. For example, studies have successfully prioritized candidate drugs for conditions like Alzheimer's disease, cancer, and COVID-19.

Another major application is the prediction of drug-target interactions (DTIs), which is crucial for understanding

a drug's mechanism of action and for identifying new therapeutic targets. This involves predicting links between drug/compound nodes and gene/protein nodes. Similarly, predicting drug-drug interactions (DDIs) is a key focus for improving patient safety by identifying potentially harmful polypharmacy side effects.

Gene-disease association prediction is also a common task, aiming to identify genes involved in the pathology of various diseases, which can lead to new diagnostic markers and therapeutic strategies. Studies have explored this for complex conditions like Autism Spectrum Disorder and various cancers.

Other significant prediction tasks include:

- **Gene/Protein Function Annotation:** Predicting the function of genes or proteins by linking them to Gene Ontology terms or pathways.
- **Synthetic Lethality Prediction:** Identifying pairs of genes whose simultaneous inactivation leads to cell death, a promising strategy in cancer therapy.
- **Protein-Protein Interaction (PPI) Prediction:** Uncovering new interactions between proteins to better understand cellular processes.
- **Adverse Drug Reaction (ADR) Prediction:** Identifying unknown side effects of drugs to enhance pharmacovigilance.
- **Regulatory Interaction Prediction:** Predicting interactions involving non-coding RNAs like lncRNAs and their target genes.

The table below maps studies to the primary biomedical prediction tasks they address.

Study	Drug-Target Interactions	Drug-Disease Associations	Gene-Disease Associations	Drug-Drug Interactions	Protein-Protein Interactions	Other (Gene function, etc.)
Yue Hu et al.		Primary				Primary (Gene function, SL, lncRNA-mRNA)
Yue Hu et al.		Primary				Primary (SL, lncRNA-gene)
Joana Vilela et al.			Primary			
Haji Gul et al.	Primary					
Tengfei Ma et al.	Primary			Primary		
Meng Wang et al.	Secondary			Primary		Primary (Pharmacological mechanisms)
Yongkang Xiao et al.		Primary				
Yahui Long et al.	Primary					Primary (Synthetic lethality)

Study	Drug-Target Interactions	Drug-Disease Associations	Gene-Disease Associations	Drug-Drug Interactions	Protein-Protein Interactions	Other (Gene function, etc.)
Zhenxiang Gao et al.		Primary				
V. Nováček et al.				Primary		
Chih-Hsu Lin et al.	Primary	Primary	Primary			
Yaqing Wang et al.	Secondary	Secondary	Secondary	Primary	Secondary	
M. Chang et al.			Primary			
M. O. Faruk et al.	Primary		Primary			Secondary (Gene function)
Nansu Zong et al.	Primary					
Lizzy Farrugia et al.				Primary		
F. Aisopos et al.	Primary					
Catarina Canastra et al.			Primary			
Kevin McCoy et al.		Primary				
Yushan Liu et al.		Primary				
Vignesh Prabhakar et al.		Primary				
Roger Tu et al.		Primary				
Sameh K. Mohamed et al.	Primary					
Gamal K. O. Crichton et al.	Primary				Primary	Secondary (Literature-Based Discovery)
A. Daowd et al. and many more...	Primary		Primary			

Computational Validation Strategies

To assess the performance of link prediction models, studies employ a range of computational validation strategies designed to simulate realistic prediction scenarios and prevent data leakage. The most common approach involves splitting the known associations (positive triples) into training, validation, and test sets. Standard split ratios like 80/10/10 or 70/20/10 are frequently reported. Cross-validation, particularly k-fold cross-validation (e.g., 5-fold or 10-fold), is also widely used to ensure the robustness of the results.

A critical component of validation is negative sampling, the process of generating non-existent links to serve as negative examples for training and evaluation. A common strategy is to randomly corrupt known positive triples by replacing the head or tail entity. However, more stringent negative sampling methods are sometimes used to create more challenging and realistic negative examples.

To address potential data leakage and evaluate generalizability, several studies adopt more rigorous splitting strategies. Temporal splits, where models are trained on data from before a certain date and tested on data published after, provide a realistic simulation of predicting future discoveries. Disjoint or "cold-start" scenarios are also used, where the test set contains drugs or diseases with no known interactions in the training set, which tests the model's ability to make predictions for novel entities. Studies have highlighted that performance can drop significantly in these more realistic settings compared to random splits, emphasizing the importance of rigorous evaluation design to avoid overestimating model performance. Some studies explicitly remove certain edge types from the KG during training to prevent trivial predictions and information leakage.

Study	Data Split Strategy	Cross-Validation Approach	Negative Sampling Method	Data Leakage Handling
Yue Hu et al.	Not mentioned	Not mentioned	Stringent negative sampling	Not mentioned
Yue Hu et al.	Train/test split	Cross-validation	Stringent negative sampling	Zero-shot disease splits to test generalization
Joana Vilela et al.	60/20/20 train/test/validation split	Not mentioned	Negative sampling used during training	Not mentioned
Haji Gul et al.	90:5:5 train/validation/test split	Not mentioned	Circumvents negative sampling by prioritizing high-density neighbors	Not mentioned
Tengfei Ma et al.	Not mentioned	10-fold cross-validation	Not mentioned	Validation on contaminated KGs to test robustness
Meng Wang et al.	Not mentioned	10-fold cross-validation	Negative sampling used to transform objective functions	Not mentioned
Chih-Hsu Lin et al.	Time-stamped splits	10-fold cross-validation; Leave-one-mode-out	Not mentioned	Temporal splits simulate future predictions

Study	Data Split Strategy	Cross-Validation Approach	Negative Sampling Method	Data Leakage Handling
Yaqing Wang et al.	7:1:2 train/validation/test split	Not mentioned	Loss function with negative sampling for multilabel prediction	Removed drug-drug edges from external KG to prevent information leakage
M. O. Faruk et al.	Chronological splits; 2024 hold-out	5-fold temporal cross-validation	Not mentioned	Leakage-free pipeline with publication-date audits and de-duplication
Nansu Zong et al.	Not mentioned	10-fold cross-validation	Negative sample selection algorithm	Studied biased learning/testing in network-based pairwise prediction
Lizzy Farrugia et al.	70/10/20 train/validation/evaluation split	Not mentioned	Random sampling of negative samples from unknown drug pairs	Cold start scenarios for drugs with no known interactions
F. Aisopos et al.	Not mentioned	10-fold cross-validation	Negative pairs generated from non-interacting drug-gene pairs	Not mentioned
Yushan Liu et al.	Random train/validation/test splits	Not mentioned	Negative sampling used for evaluation	Not mentioned
R. Çelebi et al.	Disjoint splits (drug-wise, pairwise)	10-fold traditional cross-validation	Randomly selected from unknown drug pairs	Disjoint cross-validation scheme to address evaluation biases
and many more...				

Evaluation Metrics and Benchmarking

A wide array of evaluation metrics is used to quantify the performance of link prediction models. These can be broadly categorized into ranking-based metrics and classification-based metrics. Ranking metrics are prevalent, as link prediction is often framed as a ranking problem where the model must prioritize true but missing links among a large set of candidate links. The most common ranking metrics include Mean Reciprocal Rank (MRR), Hits@k (e.g., Hits@1, Hits@3, Hits@10), and Mean Rank (MR).

Classification metrics are also widely used, especially when the task can be framed as a binary classification problem (link exists vs. does not exist). These include the Area Under the Receiver Operating Characteristic curve (AUC-ROC), the Area Under the Precision-Recall curve (AUC-PR or AUPR), F1-score, accuracy, precision, and recall.

Studies typically benchmark their proposed methods against a set of established baseline or state-of-the-art approaches. Common baselines include traditional KGE models like TransE, DistMult, ComplEx, and RotatE, as well

as more recent GNN-based models like R-GCN. The choice of benchmark datasets is also crucial for comparability. Widely used biomedical KG datasets for benchmarking include DrugBank, KEGG subsets like KEGG50k, Hetionet, and the Drug Repurposing Knowledge Graph (DRKG). However, some studies note inconsistencies in evaluation protocols across the literature, such as variations in data splits, negative sampling strategies, and the specific metrics reported, which can complicate direct comparison of results. Several studies also validate against gold standard datasets derived from sources like ClinicalTrials.gov or manually curated expert knowledge to assess the clinical relevance of predictions.

Study	Primary Metrics	Benchmark Datasets	Baseline Methods	Standardized Protocols Followed
Yue Hu et al.	MRR, Hits@k, AUPRC	PrimeKG, SynLethDB-v2.0	TransE, DistMult, RotatE, R-GCN, TxGNN, KR4SL	Not mentioned
Joana Vilela et al.	MRR, Hits@k, MR	Custom KG from DisGeNET and GO	ComplEx, DistMult, TransE	Not mentioned
Haji Gul et al.	MRR, Hits@k	KEGG50k, FB15k-237, WN18RR	ComplEx-SE, MuCo-KGC	Not mentioned
Tengfei Ma et al.	AUC-ROC, AUC-PR, F1-score	DrugBank, DrugCentral, TWOSIDES	GCN-KG, RotatE, GraIL, SNRI, AdaProp, TriModel, KGE-NFM, SumGNN	Not mentioned
Meng Wang et al.	MeanRank, Hits@k, PR curves	Bio2RDF, DDI Corpus	Tiresias, SCNN, MDDP, TransE, TransR	Not mentioned
Yongkang Xiao et al.	MRR, AUROC	KEGG50k, Hetionet, SuppKG, ADInt	Traditional KGE models, conventional GNN model	Not mentioned
M. Faruk et al.	MRR, Hits@k, AUROC	PGx-KG (custom)	Not specified, but uses a bilinear model in PyTorch Geometric	Yes, leakage-free processing pipeline with chronological splits
Lizzy Farrugia et al.	F1-score, Accuracy, Precision, Recall	DrugBank	DeepDDI, Ryu et al., Celebi et al., Karim et al.	Not mentioned
F. Aisopos et al.	Precision, Recall, F1-score	Union of DrugBank, KEGG, DGIdb, TTD	AnyBURL, SemaTyP, BLGPA, RGCN, TransE	Not mentioned
Yushan Liu et al.	MRR, Hits@k	Hetionet, FB15k-237, WN18RR	AnyBURL, TransE, DistMult, ComplEx, ConvE, RESCAL, pLogicNet	Not mentioned

Study	Primary Metrics	Benchmark Datasets	Baseline Methods	Standardized Protocols Followed
Sameh K. Mohamed et al.	AUC-ROC, AUC-PR	Yamanishi_08, DrugBank_FDA	DDR, NRLMF, KRONRLS-MKL, COSINE, BLM-NII	Not mentioned
Xiang Yue et al.	AUC, Accuracy, F1 score	CTD, NDF-RT, DrugBank, STRING	LRSSL, DeepDDI	Not mentioned
and many more...				

Performance Results and Comparative Analysis

The quantitative performance of link prediction methods varies considerably depending on the prediction task, knowledge graph, and evaluation protocol. High performance is frequently reported, with many studies achieving AUC-ROC scores above 0.90 and MRR scores exceeding 0.80 or even 0.90 in certain settings. For instance, a study on gene-disease prediction reported an MRR of 0.96 and Hits@1 of 0.93, while a DDI prediction model achieved an F1-score of 95.19%. Another framework, FuseLinker, reported an MRR of 0.969 on the KEGG50k dataset.

Comparative analyses consistently show that novel and more complex methods tend to outperform simpler baselines. For example, the path-based GNN BioPathNet outperformed shallow node embedding methods and relational GNNs. The neuro-symbolic method PoLo demonstrated performance gains of up to 27.7% for Hits@1 and 16.2% for MRR over baseline methods on a drug repurposing task. Ensemble methods have also been shown to improve performance, with one study finding that an ensemble of KGE models could achieve better results than any single model alone, and another demonstrating that a weighted ensemble surpassed standalone methods. Similarly, integrating multiple data modalities, such as text and graph structure, has been shown to yield substantial improvements, with one study reporting MRR gains of 13-36%.

However, the magnitude of these improvements is highly context-dependent. Performance can be influenced by the topological properties of the KG, the presence of noise, and the evaluation setup. Studies using more realistic validation schemes, such as temporal or cold-start splits, often report lower, albeit more generalizable, performance metrics compared to those using random splits. For example, one study reported a 20.2% average AUPRC increase in zero-shot disease splits, highlighting the challenge of predicting associations for novel entities. The top-performing approaches are often task-specific; for instance, tensor factorization methods like ComplEx and its variants have shown strong performance for polypharmacy side-effect prediction, while path-based and GNN methods excel at tasks requiring multi-hop reasoning and interpretability.

Study	Method	Prediction Task	Primary Metric	Achieved Value	Best Baseline Value	Improvement Margin
Yue Hu et al.	BioPathNet (BioKGC)	Drug-Disease	AUPRC	Not specified	Not specified	20.2% (avg)
Joana Vilela et al.	TransE (l2)	Gene-Disease	MRR	0.96	Not specified	Not specified
Haji Gul et al.	MuCoS	Drug-Target	MRR	0.84	Not specified (ComplEx-SE)	6%

Study	Method	Prediction Task	Primary Metric	Achieved Value	Best Baseline Value	Improvement Margin
V. Nováček et al.	Tensor De-composition	Polypharmacy Side-Effects	AUC-ROC	Not specified	Not specified	12%
Lizzy Farrugia et al.	ComplEx-LSTM	DDI	F1-score	95.19%	89.58% (DeepDDI)	5.61%
Yushan Liu et al.	PoLo	Drug Repurposing	Hits@1	Not specified	Not specified	27.7%
Srivamshi Pittala et al.	Not specified	Disease Gene Identification	Not specified	Not specified	Not specified	24.1%
Weichuan Wang et al.	MDistMult	Link Prediction (Antiviral Drugs)	MRR	0.244	0.083 (DistMult)	~194%
Soham Dasgupta et al.	Weighted Deep-Walk/TransE	ADE Prediction	F1-score	Not specified	Not specified	up to 5.75%
Xiaomin Liang et al.	Cascade Learning Framework	Link Prediction	Hits@10	Not specified	Not specified	8.6%
and many more...						

Biological and Experimental Validation

Beyond computational metrics, a subset of the reviewed studies undertook biological or clinical validation to confirm the real-world relevance of their predicted associations. This validation serves as a critical bridge between computational hypothesis generation and experimental verification. The most common form of non-computational validation is literature-based analysis, where researchers manually search scientific publications (e.g., on PubMed) to find evidence supporting a novel predicted link. For instance, novel drug indications for acute lymphoblastic leukemia and Alzheimer's disease were confirmed through literature reviews, and a predicted link between the gene VDR and prostate cancer was substantiated by published research.

Another approach involves checking predictions against independent, curated databases that were not used in the model's training. This can include clinical trial databases like ClinicalTrials.gov, pharmacovigilance databases like EudraVigilance, or comprehensive interaction databases like STITCH, OpenTargets, and DrugBank. In one study, 14 out of 15 top-ranked predicted gene-disease associations were supported by evidence from the TINX database. Another found that 60% of its predicted DDIs were present in a newer, updated version of DrugBank.

Expert review by clinicians or domain specialists is also used to assess the plausibility of predictions. One study reported that physicians deemed 33% of false positives for treatment relations to be plausible, with many identified as potentially novel findings.

Direct experimental validation is the most rigorous but least common form of validation reported in these computational studies. One notable example involved a patient-derived in-vitro assay for Rheumatoid Arthritis, which was used to test predictions from the Rosalind method and successfully identified five promising genes, one of which

was previously unexplored in the disease context. Another study experimentally validated predictions of previously unknown kinase-substrate phosphorylations using kinase assays. These instances, while rare, demonstrate the potential of computational models to guide targeted and successful wet-lab experiments.

Study	Validation Type	# Predictions Validated	Validation Success Rate	Notable Novel Discoveries
Yue Hu et al.	Expert review, Clinical trials, Literature	Not specified	Not specified	Novel drug indications for ALL and Alzheimer's; new synthetic lethality gene pairs
Yushan Liu et al.	Clinical trials database (ClinicalTrials.gov)	Not specified	Not specified	Potential new uses for sorafenib (pancreatic cancer) and ergocalciferol (hypertension)
Sameh K. Mohamed et al.	Literature review	Top 10	70% or more	High ranking of recently discovered drug-target interactions
S. Paliwal et al.	Experimental (in-vitro assay)	Not specified	5 promising genes identified	One gene unexplored in Rheumatoid Arthritis was identified as promising
D. Bean et al.	Electronic health records, Pharmacovigilance database	Not specified	Significantly more frequent than random models	Predictions for akathisia and pulmonary embolism
Gokhan Bakal et al.	Expert (physician) review	Not specified	33% of false positives deemed plausible	68% of physician-validated predictions were potentially novel findings
V. Nováček et al.	Experimental (kinase assays)	Not specified	Not specified	Confirmed new substrates for LATS1 and AKT1 kinases
Nansu Zong et al.	Database (DrugBank 5.1.0)	75	Not specified	75 novel drug-target associations predicted and validated

Study	Validation Type	# Predictions Validated	Validation Success Rate	Notable Novel Discoveries
Daniel N. Sosa et al.	Database, Network proximity analysis	30	Not specified	30 novel repurposing candidates for rare diseases, many corroborated by external data
Saatviga Sudhahar et al.	Experimental (transcriptional changes)	Not specified	Strong correlation observed	Ibuprofen linked to Fragile X syndrome via cAMP/cGMP pathways
and many more...				

Technical Implementation and Reproducibility

Technical implementation details vary substantially across studies, reflecting the diversity of computational approaches. For KGE models, the embedding dimension is a key hyperparameter, typically ranging from 50 to 512, with values of 100, 128, or 200 being common. GNN architectures often specify the number of layers, with single-layer and two-layer GCNs being reported, as well as deeper architectures with up to six layers.

Training procedures commonly involve optimizers like Adam or AdamW, with learning rates tuned in ranges such as $1e-5$ to $1e-2$. The number of training epochs can range from a few dozen to several hundred, often with early stopping criteria to prevent overfitting. Batch sizes are also specified, ranging from 16 to 1024. Computational requirements can be significant, with many studies reporting the use of high-performance computing clusters equipped with GPUs like the NVIDIA V100 or A100. Training times can range from under a minute for highly optimized methods to several hours.

In terms of reproducibility, there is a positive trend towards making code and data available. A significant number of studies provide links to GitHub repositories containing their source code and datasets. Popular software frameworks for implementation include PyTorch, TensorFlow, and specialized libraries for graph machine learning and KGE, such as PyTorch Geometric, DGL-KE, OpenKE, and PyKEEN.

Handling the heterogeneity of biomedical KGs is a common challenge. Approaches include using multi-relational GNNs like R-GCN, employing models that can inherently handle multiple relation types (e.g., ComplEx, RotatE), integrating different data modalities through fusion techniques, or using meta-paths to navigate the diverse relationships in the graph.

Study	Embedding Dimensions	Key Hyperparameters	Computational Resources	Training Time	Code/Data Availability
Yue Hu et al.	32	layers: 6, lr: 0.005, batch_size: 8-64, epochs: 8-10	Not specified	Not specified	Not specified

Study	Embedding Dimensions	Key Hyperparameters	Computational Resources	Training Time	Code/Data Availability
Joana Vilela et al.	Varied (tuned)	max_step, hidden_dim, neg_sample_size, batch_size, lr tuned with Optuna	Not specified	Not specified	No
Haji Gul et al.	Not specified	epochs: 50, lr: 5e-5, batch_size: 16	NVIDIA GeForce RTX 3090 GPU (24 GB)	Not specified	No
Tengfei Ma et al.	32-512	lr: 1e-4 to 1e-1, subgraph_size: 1-4 hop, weight λ : 0.1-1	Linux server, 2x RTX 2080Ti GPUs	Not specified	Not specified
Wen Tao et al.	256	lr: 0.0001, max_epochs: 200	NVIDIA GeForce RTX 2080 Ti GPU	Not specified	Yes
Daniel N. Sosa et al.	50 (TransE), 200 (ComplEx/RotatE)	epochs: 500, lr: 0.02, negative_samples: 50	Not specified	Not specified	Yes
Asan Agibetov et al. and many more...	50	Not specified	Not specified	< 1 minute	No

Synthesis of Methods and Validation Practices

An integrative synthesis of the reviewed literature reveals several emerging patterns and best practices in the computational prediction of biomedical associations. Methodologically, there is a clear trend moving from traditional KGE models (e.g., TransE, DistMult) towards more sophisticated architectures that leverage graph neural networks (GNNs), path-based reasoning, and the integration of multimodal data, particularly textual information from large language models (LLMs). GNNs, especially those incorporating attention mechanisms or designed to handle multi-relational, heterogeneous data, are consistently reported as top-performing approaches for capturing complex topological patterns in biomedical KGs. Hybrid and ensemble methods that combine the strengths of different approaches—such as combining the global structural awareness of KGE with the local neighborhood aggregation of GNNs, or integrating path-based and embedding-based predictions—are also gaining prominence as a strategy to boost performance and confidence. Furthermore, neuro-symbolic methods that incorporate logical rules or domain knowledge into neural models are emerging as a powerful way to enhance reasoning and provide interpretability, a crucial aspect for biomedical applications.

In terms of validation, a consensus is forming around the need for more rigorous and realistic evaluation protocols. While simple random train-test splits and k-fold cross-validation remain standard practice, there is a growing acknowledgment of their limitations, particularly the risk of data leakage and overly optimistic performance estimates.

Consequently, more studies are adopting temporal splits and "cold-start" or disjoint validation scenarios as a more robust benchmark for assessing a model's ability to generalize to truly novel entities and future discoveries. The choice of evaluation metrics is also maturing, with a combined use of ranking metrics (MRR, Hits@k) to assess prioritization ability and classification metrics (AUC-ROC, AUC-PR) to evaluate discrimination performance becoming standard.

However, a significant gap persists between computational validation and biological verification. While many studies demonstrate high computational performance, only a fraction proceed to validate their novel predictions through literature review, independent database confirmation, expert review, or, most rarely, direct experimental testing. This gap highlights a critical challenge for the field: translating high computational scores into biologically meaningful and actionable discoveries. The most effective strategies appear to be those that not only achieve high performance on rigorous computational benchmarks but also provide interpretable outputs (e.g., through attention weights or reasoning paths) that can guide and facilitate subsequent biological validation.

Methodological Limitations and Research Gaps

Despite significant progress, the field of biomedical knowledge graph link prediction faces several methodological limitations and underexplored research areas. A primary challenge is the issue of **evaluation bias and lack of standardization**. Many studies rely on random data splits, which can lead to data leakage and artificially inflated performance metrics, as models may learn simple heuristics rather than complex biological patterns. The inconsistency in evaluation protocols, including different negative sampling strategies, benchmark datasets, and reported metrics, makes it difficult to compare methods across studies reliably. This points to a need for standardized, challenging benchmarks that incorporate realistic scenarios like temporal validation and cold-start prediction for novel entities.

Limited external and biological validation remains a major gap. The vast majority of studies stop at computational validation, and there is a scarcity of research that follows through with literature-based confirmation, expert review, or experimental verification of novel predictions. This gap between computational prediction and biological reality hinders the translation of these methods into practical tools for biomedical discovery. More integrated, interdisciplinary efforts are needed to bridge this divide.

Reproducibility is another concern. While the trend towards publishing code and data is positive, many studies still do not provide sufficient implementation details, hyperparameter settings, or access to their resources, impeding independent verification and extension of their work.

From a methodological standpoint, many current algorithms still struggle with several intrinsic challenges of biomedical KGs. Handling the **extreme sparsity and incompleteness** of these graphs is a persistent problem. While some methods attempt to address this through techniques like denoising or leveraging unlabeled information, it remains an open area of research. Similarly, effectively modeling the complex, multi-relational, and heterogeneous nature of biomedical data continues to be a focus of innovation, with a need for methods that can better integrate diverse data modalities (e.g., genomics, proteomics, clinical data, and text) in a cohesive framework.

Finally, there are several underexplored biomedical applications. While drug-target, drug-disease, and gene-disease predictions are well-studied, other areas such as predicting interactions involving non-coding RNAs, understanding the molecular basis of adverse drug events, and predicting the effects of genetic variants on interactions are less explored. Methodologically, there are opportunities for developing more powerful neuro-symbolic models that can better integrate domain knowledge and provide more robust explanations, as well as for creating inductive models capable of making predictions for entities not seen during training, which is crucial for real-world applications.

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